Wilfrid Laurier University

PC141 Lab Manual

Estimation, Bounding, and Order of Magnitude Calculations

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2026

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1 Purpose

In this lab we will begin to build the foundation for working with **measurement uncertainty**. Before learning formal methods, it is essential to practice concepts such as *estimation*, *bounding*, and *order-of-magnitude reasoning*. Especially when it is useful in situations where quantities cannot be measured directly. These skills help us decide whether a result is reasonable and how reliable it might be.

These skills form the first step toward a systematic treatment of uncertainty, which you will develop further in later labs. These first steps mirror how scientists approach real problems: starting with rough estimates, bounding possibilities, and then refining to quantified uncertainties.

2 Introduction

Although science is rooted in precise measurement, scientists and engineers often begin with approximate estimates. These serve as a 'sanity check'—a way to anticipate results, choose appropriate instruments, and identify mistakes early (e.g., wrong units or unrealistic values). This exercise about how to move confidently between rough intuition and systematic measurement.

3 Theory and Procedure

3.1 Estimation

In many experiments, you will face situations where quantities cannot be measured immediately or directly. In these cases, it is valuable to begin with an *estimate*—a reasoned approximation of the result you expect. Estimation helps you decide what instruments or methods are appropriate and provides a reference point against which later measurements can be compared.

What matters most is not whether your estimate is exact, but whether you have a clear idea of how far off it *could* be. Even a rough estimate can be very useful when it

helps you detect mistakes, choose equipment, or check the plausibility of results.

Mini rule-of-thumb: Estimates within a factor of 2–3 are typical of reasonable human judgment. A difference of a full order of magnitude (a factor of 10) often signals a unit error or faulty reasoning.

3.2 Bounding

A natural companion to estimation is **bounding**. Bounding a quantity means identifying values that it *cannot* reasonably exceed or fall below. An **upper bound** is a value greater than the true result, while a **lower bound** is a value smaller than the true result. Together, these bounds form a safe "bracket" around the unknown value.

3.2.1 Picking Realistic Bounds

Reasonable bounds can often be set by comparison with known quantities. For example:

- To estimate a person's height, you might compare them with friends or family members of known height. A taller person gives an upper bound; a shorter person gives a lower bound.
- To estimate the mass of an unfamiliar object, you can compare it to objects you know well. Something clearly heavier provides an upper bound, while something lighter provides a lower bound. Wide bounds are safe but not useful. Narrower bounds make your estimate more powerful.

3.2.2 Range of Possible Values

Once upper and lower bounds are set, the **range of possible values** can be defined as the difference between them. This range is a first step toward the idea of *uncertainty*: the narrower the range, the more useful the estimate.

3.3 Familiar Comparisons

When the unknown quantity is similar to something you already know, comparison is one of the most effective tools for estimation. If you can bracket the unknown value between a higher and a lower bound, you can usually place your estimate somewhere in between with reasonable confidence.

For example, if you know the mass of one object is smaller than your upper bound and larger than your lower bound, then your estimate should reflect a value between those two reference points.

3.3.1 Illustration of Comparison

To illustrate, we'll try to make a few simple estimates. The first question we want to answer is:

IQ1: How tall is your TA?

1. Identify a person who is shorter than the IA (this person can be one of your group members or a celebrity). Record that person's height. This quantity is a lower bound for the IA's height.

2. Identify a person who is taller than the IA (this person can be one of your group members or a celebrity). Record that person's height. This quantity is an upper bound for the IA's height.

3. Estimate the IA's height according to the two known heights and explain your reasoning.

4. Adjust the bounds you would place on the IA's height so that there is a smaller range of values. Are these bounds the same as a) and b)? If not, explain how you adjusted (and estimated) the bounds. (i.e. if you saw someone about his/her height commit a crime, what range of heights would you give to investigators so that it would be of use in identifying suspects? The range of values has to be very small for it to be of good use!!!)



The goal when making estimates is to try and make them "safe" but "useful"; i.e. you are *pretty sure* about lower and upper bounds on your estimate, but the bounds need to be close enough together to make the estimate usable.

IT1: Fill in Table 1.

Based on comparisons, fill in Table 1. Follow the steps of IQ1 for each of the quantities. Choose a reasonable estimate and bounds (bounds should lead to a small range of values!) for each of the following:

- IAs height (from above)
- mass of a block of wood
- volume of liquid

Scientists face both familiar and unfamiliar problems. Familiar cases build intuition; less familiar cases require creative comparisons and sometimes fuzzier bounds.

3.4 Less Familiar Comparison

In some cases, it is not easy to find a familiar object to compare with the quantity of interest. When there is nothing clearly similar, your estimate will naturally become less precise and the bounds you choose will be more uncertain.

For example, it is difficult to set upper and lower bounds for the height of the Eiffel Tower. Unlike comparing a person's height to family or friends, there is no everyday reference object that is truly comparable. In such situations, estimates and bounds must be broader and more approximate, yet they are still useful.

IT2: Fill in Table 1. Using this idea, fill in Table 2 by suggesting bounds and then an estimate. Adjust your bounds to make the range as small as is reasonably possible, and record a final estimate for each of the following:

- the height of Conrad Hall (in metres),
- the length of Conrad Hall (in metres),
- the mass of a laboratory table (in kilograms)
  (Please do not attempt to lift the table!)

IQ2: For one of the quantities in Table 2:

1. State your upper and lower bounds, and explain how you chose them.
2. Give your estimate and describe your reasoning.
3. Compare the size of your range with those from the more familiar comparisons above. Was it larger? Why or why not?

3.5 Order of Magnitude Calculations

In physics labs, **order of magnitude (OM)** calculations are often done early, using preliminary or estimated data, to check whether results seem reasonable. They serve



Figure 1: Conrad Hall, for comparison

as a quick "sanity check" before spending time on precise measurements.

Order of magnitude refers to the *power of ten* that best describes a quantity. Raising a number by one order of magnitude multiplies it by 10; lowering it divides it by 10. For example:

- 200 and 600 are the same order of magnitude (hundreds, or 10^2).
- 0.02 is one order of magnitude smaller than 0.2.
- A city with 1,000,000 people (10^6) is three orders of magnitude larger than a town of 1,000 people (10^3).

Since an order of magnitude calculation is intended to be "within a factor of ten," you should be reasonably confident that:

- Ten times your value is clearly too large,
- One-tenth of your value is clearly too small.

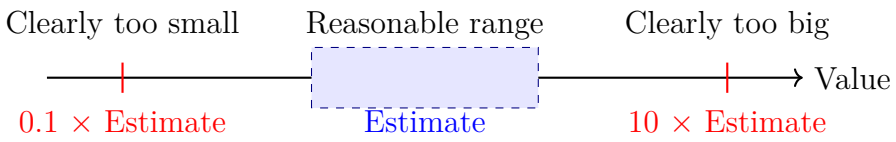


Figure 2: Order-of-magnitude self-check: An estimate should lie in a range where 10× is clearly too big and 0.1× is clearly too small.

Why Order of Magnitude Matters

- **Error detection:** Using the wrong unit (e.g., cm instead of m) will often shift your result by several orders of magnitude, making the mistake obvious.
- **Quick checks:** Scientists use OM reasoning to judge whether a result is physically realistic.
- **Decision making:** OM estimates guide choices of instrument sensitivity or experimental method.

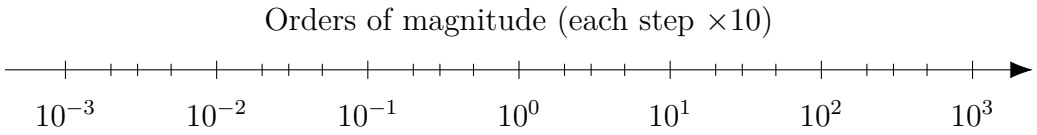


Figure 3: Orders of magnitude illustrated on a logarithmic number line. Each major tick is a factor of 10; minor ticks show common intermediate factors (×2, ×3, ×5).

3.5.1 Example: How many gold parking spaces are there at Laurier?

Estimation problems in science often ask us to reason from limited data. The goal is not to get an exact number but to reach a reasonable *order-of-magnitude* estimate.

Let us try this with the number of *gold parking spaces* (faculty and staff only) on campus.

1. Gold lots are reserved for faculty and staff, so the number of spots must be smaller than the total number of employees at Laurier, since many walk, take the bus, or carpool.
2. There are several lots of different sizes. A good strategy is to estimate the average size of a lot and multiply by the number of lots.

From observation:

- The lot in front of the Science Building has about 50 spaces (roughly 25 per side).
- Suppose there are about 10 similar lots across campus.

If S is the average number of spaces per lot and L is the number of lots, then

$$N = S \times L$$

so with $S \approx 50$ and $L \approx 10$,

$$N \approx 50 \times 10 = 500.$$

Does this make sense?

- *More than 50?* Yes—since one lot already has about 50.
- *Less than 5000?* Laurier has about 10,000 students. For 5000 gold spots to exist, there would need to be one faculty or staff member for every two students. In reality, the ratio is closer to 1 staff for every 10 students, so 5000 is much too high.

Thus, an estimate of about **500 gold spots** is reasonable.

One of the important features of order-of-magnitude reasoning is that it is often **self-correcting**: you may overestimate some parts and underestimate others, but the errors tend to balance out, leaving you close to the right order of magnitude.

3.5.2 Illustration of Order of Magnitude Calculation

Any time you use an *estimated* quantity in a calculation, you are performing an **order of magnitude (OM) calculation**. The purpose is not to obtain an exact answer, but to check whether the result falls in a reasonable range.

The *order of magnitude* of a quantity refers to the power of ten that best describes it. For example:

- The height of a typical building might be written as $\sim 10^1$ metres (tens of metres).
- The length of an entire campus might be $\sim 10^2$ metres (hundreds of metres).

Thus, saying that the length of the Science Building is

- "about 100 metres," or
- "between 50 and 200 metres,"

are both acceptable order-of-magnitude estimates. What matters is that the result lands in the correct *ballpark*.

3.5.3 When are Order of Magnitude Calculations used?

Order of magnitude reasoning is common in science and engineering:

- **Before experiments:** to anticipate what range of values to expect, and to decide which instruments or methods are suitable.
- **During data collection:** to check whether preliminary results make sense in real time.
- **Afterward:** to identify errors, such as unit mistakes, which often shift results by entire orders of magnitude.

Key idea: Any calculation involving at least one estimated quantity is an order-of-magnitude calculation. The more estimates involved, the wider the range of possible answers, but the result should still be close to the "real" answer. Errors often balance out, making OM estimates *self-correcting*.

3.6 Logarithmic Scale

You are not expected to be fluent with logarithms yet — the goal is to recognize relative scale, not perform algebra.

One of the most effective ways to visualize orders of magnitude is with a **logarithmic scale**.

Unlike a linear scale, where equal spacing means equal differences, a logarithmic scale increases by constant factors, usually powers of ten. Each interval is called a **cycle**, covering one power of ten. For example:

- from 1 to 10 is one cycle,
- from 10 to 100 is the next cycle,
- from 0.1 to 1 is also one cycle.

Key idea: If two numbers fall within the same cycle of a logarithmic scale, they are within one order of magnitude of each other. This is why logarithmic plots are especially useful for comparing estimates.

Figures 4–6 show examples of logarithmic scales with different ranges.

If two numbers fall within the same cycle of a logarithmic scale, they are within one order of magnitude of each other. This is why log scales are so useful for comparing estimates.

On a logarithmic scale, estimates that fall within the same cycle share the same order of magnitude. Numerical differences are acceptable; falling in a different cycle indicates a wrong order of magnitude.

IT3: For one of the quantities estimated

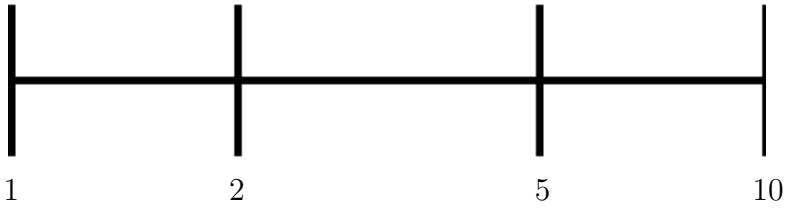


Figure 4: Logarithmic scale (single cycle from 1 to 10).

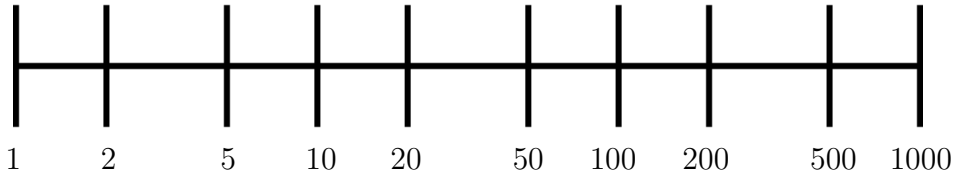


Figure 5: Three-cycle logarithmic scale (1 to 1000).

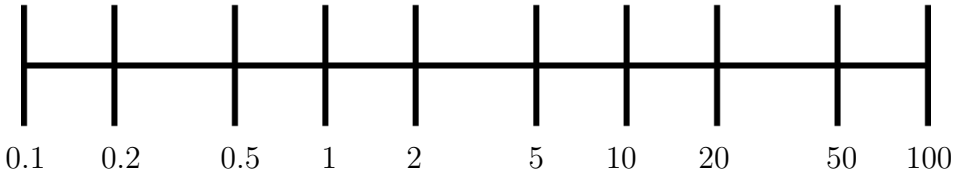


Figure 6: Logarithmic scale including numbers less than one (0.1 to 100).

in Table 2, use the logarithmic scale of Figure 7 and clearly label and name another group' estimate on it. (You first must take the logarithm of the estimate before plotting it on the logarithmic scale! For example, if Jack and Jill estimated 16m for the height of Conrad Hall. Take the logarithm of 16 which is 1.2 (unit-less). Write "Jack and Jill - 1.2 above the logarithmic scale and draw an arrow to the 1.2 point on the scale). Also write a title for Figure 7.

There are many things which we perceive on a logarithmic scale (such as the volume of music and pH of a chemical solution). *What is another example?*

IQ3: Use Figure 7 to answer the following:

1. Do all of the estimates in Figure 7 for each quantity fall within a single cycle of the scale?
2. Which cycle?
3. Explain what falling within one cycle or not falling within one cycle indicates about the data.

For any question, always include a sentence or two of explanation, even if a single word answer such as 'yes' or 'no' is possible (like: What did I compare it to? Why is this value reasonable?).

3.7 Uncertainties

Estimation and bounding naturally lead us to the idea of **uncertainty**. An uncertainty tells us how confident we are about a measured or estimated value, expressed as a central value ± some range. Instead of always writing separate lower and upper bounds, we compress this information into a single statement.

From Bounds to Uncertainty If you can state a credible lower bound L and upper bound U, a practical estimate and uncertainty are:

$$\hat{x} = \frac{L+U}{2}, \qquad \Delta x \approx \tfrac{1}{2}(U-L), \qquad \text{report as } x = \hat{x} \pm \Delta x.$$

- **Uncertainty** = the range we expect, based on limits of measurement or estimation.
- **Error** = the actual difference between a measured value and the true value (unknown in most real experiments).

Even if your value equals a "known" reference, you must still report an uncertainty. Because uncertainty reflects how the value was obtained, not how close it happens to be.

IT4: Using the information from Table 3:

1. For each quantity, compute the midpoint $\hat{x}$ and uncertainty $\Delta x \approx \frac{1}{2}(U-L)$.

2. If your best estimate is not midway between the bounds, either (i) choose $\Delta x = \max\{U - E,\ E - L\}$ or (ii) adjust one bound to make the interval symmetric. Here, E denotes your best estimate of the quantity.
3. Record each **experimental statement** in the form $x = \hat{x} \pm \Delta x$ with correct units.

Notation

The uncertainty in a symbol is indicated with the Greek letter Δ. If a mass is m, then Δm denotes the *uncertainty in the mass*, and you would write $m \pm \Delta m$. Uncertainties are always reported as *positive* numbers.

IQ4: Using Table 3:

1. Did any of your estimates *not* fall midway between your upper and lower bounds?
2. If so, how did you choose your uncertainty? If not, how would you choose your uncertainty if your estimate did not fall exactly midway between your upper and lower bound?

Post Lab Question

A post-lab question for this lab is posted on myLS.

Recap

By the end of this exercise, you should understand the following terms and ideas:

- **Estimate:** a rough calculation of a quantity when direct measurement is not possible.
- **Bound:** upper and lower limits that frame where a quantity is likely to fall.
- **Range:** the difference between the upper and lower bounds.
- **Order of magnitude:** the scale or power of ten that best describes the size of a quantity.
- **Logarithmic scale:** a nonlinear scale used to represent values that span several orders of magnitude.
- **Uncertainty:** the confidence interval around an estimate, expressed as ± some value.
- **Agreement:** when two values with uncertainties overlap in their ranges, suggesting they may represent the same true value.

4 Summary

Item	Number	Received	Weight (%)
Pre-lab Questions	0	_____	0
In-lab Questions	0	_____	40
Post-lab Questions	0	_____	0
Pre-lab Tasks	0	_____	0
In-lab Tasks	0	_____	40
Post-lab Tasks	0	_____	20

5 Template

My name: _____

My partner's name: _____

My lab section: _____

My lab IA name: _____

Today's date: _____

☐ I confirm that I have brought a printed copy of the lab manual.

Read and analyze In-lab Tasks before filling the tables.

Table 1: Estimates for Task 1

Quantity	**Estimate**	**Units**	**Upper bound**	**Lower bound**
IA's height				
Mass of a block of wood				
Volume of liquid				

Table 2: Estimates for Task 2

Quantity	**Estimate**	**Units**	**Upper bound**	**Lower bound**
Height of Conrad Hall		m		
Length of Conrad Hall		m		
Mass of lab table		kg		

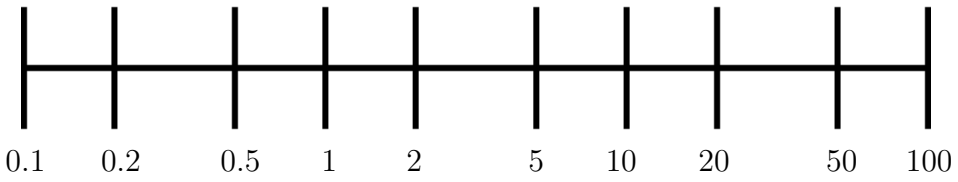


Figure 7: Logarithmic Scale for Task 3

Table 3: Determining Uncertainties for Task 4

Quantity	**Estimate**	**Units**	**Upper bound**	**Lower bound**	**Uncertainty**
My height					
Mass of a block of wood					
Volume of liquid					

Appendix A: Reference Material

A.1 Significant Figures (SigFigs)

Why they matter. Significant figures reflect the precision of a measurement and keep reported results consistent with the quality of the data.

Core rules

- All nonzero digits are significant (e.g., 123.4 has 4 sig figs).
- Zeros between nonzero digits are significant (e.g., 1002 has 4 sig figs).
- Leading zeros are *not* significant (e.g., 0.00450 has 3 sig figs).
- Trailing zeros are significant *if* a decimal point is present (e.g., 45.00 has 4 sig figs; 4500 has 2 sig figs unless written as 4.500×10^3).

Operations

- **Multiplication/Division:** keep the same number of sig figs as the factor with the fewest sig figs.
  Example: $(2.6\text{ cm}) \times (3.14\text{ cm}) = 8.164\text{ cm}^2 \to 8.16\text{ cm}^2$ (3 sig figs).
- **Addition/Subtraction:** round to the least precise *decimal place*.
  Example: $12.30 + 0.456 = 12.756 \to 12.76$ (hundredths place).

Worked example You measure a block's edges as 12.3 cm, 3.40 cm, and 0.85 cm. Volume:

$$V = (12.3)(3.40)(0.85) = 35.511\text{ cm}^3 \;\to\; \boxed{35.5\text{ cm}^3}$$

(3 sig figs overall; the limiting factor is 0.85 with 2 sig figs? No—0.85 has *two* sig figs, so strictly you should report 35 cm^3. If your instrument actually supports *three* sig figs for each edge, keep 3 sig figs. Match this to your instrument's markings.)

Common pitfall Writing 4500 without context is ambiguous. Prefer 4.5×10^3 (2 sig figs), 4.50×10^3 (3), or 4.500×10^3 (4).

A.2 Scientific Notation (Standard Form)

Definition. Write numbers as $N = a \times 10^n$ where $1 \leq a < 10$ and $n \in \mathbb{Z}$.

Why use it?

- Removes ambiguity about significant figures.
- Makes orders of magnitude clear and arithmetic simpler.

Examples

$$0.00023 = 2.3 \times 10^{-4}, \qquad 1{,}200{,}000 = 1.2 \times 10^6.$$

Common pitfalls

- Writing 12×10^3 instead of 1.2×10^4. Keep a in $[1, 10)$.
- Dropping trailing zeros that *are* significant (use scientific notation to preserve them).

A.6 Quick Pre-Submission Checklist

- Units on all quantities; scales reasonable.
- Results reported with appropriate sig figs.
- Where applicable, an uncertainty statement $x = \hat{x} \pm \Delta x$.
- If an OOM check was possible, briefly note it.